

JOBBEACON

Product Requirements Document

Autonomous Vacancy Discovery, Trust, Company Intelligence & Application Platform

Version 3.0

Document status	Draft for Founder Approval
Owner	JobBeacon Product
Primary markets	India, United States, Europe
Currencies	INR, USD, EUR
Prepared date	24 July 2026
Approval required	Product scope, sourcing policy, trust model and rollout

Core operating principle

After a candidate confirms verified facts and selects target roles, JobBeacon independently discovers, verifies and applies to eligible vacancies. Routine candidate assistance is not required.

Document Control

Field	Requirement
Authority	This PRD is the product-level source of truth for vacancy sourcing, trust, company intelligence, moderation and autonomous application behavior.
Relationship to existing documents	Extends the JobBeacon Master Product Specification v2.1 and the Frontend/UI-UX v2.1 documents.
Change control	Material changes require founder approval and a version increment.
Implementation freedom	Engineering may choose implementation details but may not weaken the required founder-visible outcome, source policy or safety controls.
Evidence standard	Every production-ready claim requires exact-head CI, deployment provenance, security evidence and founder acceptance.

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1. Executive Summary

JobBeacon is a candidate-authorized autonomous job-application platform. The candidate creates and verifies an account, uploads a resume, confirms extracted facts, and selects the roles they want to pursue. From that point, JobBeacon owns the operational workflow: vacancy discovery, source and employer verification, duplicate and freshness checks, eligibility analysis, truthful ATS resume preparation, application submission, evidence capture, retry and response tracking.

The candidate is not expected to browse a job board, shortlist vacancies, approve each opportunity, configure a mandatory campaign wizard, or click Apply repeatedly. Candidate interruption is limited to unavoidable events such as CAPTCHA, OTP, unknown legal or sensitive questions, unsupported assessments, or missing verified information.

Product promise

Select the roles once. JobBeacon continuously discovers trustworthy opportunities, verifies them, applies automatically when safe and supported, and shows evidence of what happened.

This PRD also defines JobBeacon's fake-job moderation system, company profiles, work-life-balance and compensation intelligence, vacancy-source policy, API strategy, legal restrictions, human review, employer appeals, and the data and system architecture required to support global operation in INR, USD and EUR.

2. Product Vision and Operating Model

2.1 Vision

Make verified, trustworthy and evidence-backed job applications happen automatically for candidates, without forcing them to become full-time job-search operators.

2.2 Authoritative operating model

1. Candidate registers, verifies identity and secures the account.
2. Candidate uploads a resume and confirms or corrects extracted facts.
3. JobBeacon builds a verified master candidate profile.
4. JobBeacon suggests primary, strong, related and stretch roles.
5. Candidate selects target roles and authorizes automation.
6. JobBeacon continuously discovers vacancies from allowed sources.
7. Each vacancy is normalized, deduplicated, freshness-checked and verified.
8. Company, source, eligibility, trust and scam-risk checks run automatically.
9. JobBeacon generates a truthful role-specific ATS resume from confirmed facts.
10. JobBeacon submits eligible applications automatically through supported channels.
11. Every decision and submission attempt receives evidence and audit history.
12. JobBeacon tracks replies, interviews and action-required items.
13. Candidate is interrupted only when automation cannot safely continue.

2.3 Candidate control without candidate operation

Candidate controls	JobBeacon operates
Confirm or correct facts	Vacancy discovery and source monitoring
Select or change target roles	Company and vacancy verification
Set global exclusions or hard constraints	Eligibility, trust and scam checks
Pause, resume or stop	ATS resume selection/generation
Answer unavoidable sensitive questions	Application submission and retry
Review activity and evidence	Response tracking and classification

3. Problem Statement

Candidates face a fragmented and repetitive job-search process. Vacancies are duplicated across sources, many are stale or misleading, employer quality is unclear, salary information is inconsistent, application forms are repetitive, and major platforms restrict scraping or automation. Traditional job boards solve discovery but still require candidates to research, assess, customize and apply manually.

Employers and legitimate job sources also face spam applications, bot abuse, inaccurate candidate data and untrusted aggregators. JobBeacon must create automation that is legally permitted, technically reliable, candidate-authorized and explainable to employers and candidates.

3.1 Core problems

- Vacancy authenticity is unknown or difficult to verify.
- Company quality cannot be represented by one generic rating.
- Salary signals come from sources with different confidence levels.
- Candidates waste time on duplicate, expired, ineligible or scam vacancies.
- No universal job API or universal application API exists.
- Major platforms may prohibit scraping or automated application behavior.
- Application forms vary and may contain sensitive, legal or unsupported questions.
- Automation can fabricate or overstate candidate facts unless strictly controlled.
- Moderation decisions require evidence, appeals and human oversight.

4. Goals and Non-Goals

4.1 Goals

- Enable candidate-authorized automatic applications after verified role selection.
- Discover vacancies only from sources whose access and use policy is known.
- Block or quarantine fake, suspicious, stale, duplicated or unauthorized vacancies.
- Provide explainable vacancy trust, company verification and application evidence.
- Provide separate company dimensions for work-life balance, compensation, management, culture, career growth, job security and candidate experience.
- Use licensed APIs, official ATS endpoints, employer feeds and permitted career pages rather than indiscriminate scraping.

- Escalate only unavoidable candidate actions.
- Support India, United States and Europe with regional currency and policy differences.
- Preserve candidate privacy, tenant isolation and auditability.
- Create a moderator and employer-appeal workflow that is fair and traceable.

4.2 Non-goals

- Universal automation across every website regardless of terms or technical restrictions.
- LinkedIn or Indeed scraping without explicit written authorization.
- Bypassing CAPTCHA, paywalls, login restrictions, robots controls or anti-bot protections.
- Fabricating company ratings, salaries, candidate qualifications or job facts.
- Publishing low-sample employee ratings as reliable scores.
- Allowing employers to suppress legitimate negative reviews.
- Requiring candidate approval for every vacancy or application.
- Using production candidate data in demos or model training without explicit consent.

5. Users and Stakeholders

Persona	Primary needs	Critical permissions
Candidate	Select roles once, receive safe automatic applications, understand evidence and action-required items.	Own profile, pause/resume/stop, change roles, report jobs, review evidence.
Moderator	Review suspicious vacancies and company claims efficiently.	View evidence, flag/block/clear, request information, handle appeals.
Employer representative	Claim company profile, correct objective facts, provide authorized feeds and application integrations.	Submit verification evidence, respond to reviews, appeal moderation.
Admin/Trust lead	Manage source policies, scoring rules, thresholds, SLAs and escalations.	Configure source access, risk rules, moderation policy and audit review.
Operations/support	Resolve candidate issues and action-required failures.	View redacted traces and retry workflows; no unrestricted candidate secrets.
Compliance/security	Ensure source, privacy and application behavior remains lawful and secure.	Policy review, access logs, incident response and retention controls.

6. End-to-End Candidate Journey

Stage	Candidate experience	System behavior
1. Identity	Register, verify email, login and enroll MFA where required.	Create secure account, session and audit trail.
2. Resume	Upload PDF/DOCX and review extracted facts.	Validate, store privately, parse and preserve provenance.
3. Role selection	Review primary/strong/related/stretch roles and select targets.	Resolve canonical role taxonomy and evidence.
4. Automation authorization	Confirm that JobBeacon may operate for selected roles.	Activate autonomous operation using stored preferences and safety limits.

Stage	Candidate experience	System behavior
5. Discovery	Monitor opportunities; no manual shortlist required.	Ingest, normalize, deduplicate and recheck vacancies continuously.
6. Trust and eligibility	See blocked or verified reasons when desired.	Verify source, company, vacancy, eligibility, trust and scam risk.
7. Application	No routine action.	Generate truthful ATS material and apply through supported routes.
8. Exceptions	Answer only unavoidable action-required items.	Pause one application, preserve state and resume after response.
9. Outcomes	Review submissions, replies and interviews.	Track mailbox events, classify responses and update analytics.

7. Product Principles

Principle	Requirement
Verified facts only	No employer, date, skill, education, project, certification or metric may be invented.
Autonomy after role selection	Routine candidate assistance ends after verified role selection and authorization.
Source-policy first	A worker may not access, store, display or apply through a source until policy capabilities are explicitly recorded.
Trust before matching	Vacancy verification and scam checks occur before automatic application.
Separate dimensions	Company legitimacy, employee experience, compensation and candidate experience remain separate.
Hard blocks override scores	Payment requests, phishing or domain impersonation block a vacancy regardless of aggregate score.
Evidence over claims	Every important decision has reason codes, timestamps and source evidence.
Human review for ambiguity	Automation quarantines uncertain cases rather than guessing.
Candidate stop control	Pause, resume and stop must be available at all times.
Exact deployment proof	Production claims require CI, artifact and deployment SHA parity.

8. Functional Scope

Capability	Release scope	Notes
Authentication and verified profile	Required	Email/password, verification, sessions, reset, MFA/AAL2, RLS and private storage.
Role intelligence	Required	Primary, strong, related, stretch and

Capability	Release scope	Notes
		manual canonical role selection.
Vacancy source registry	Required	Capabilities, terms, credentials, rate limits, attribution and legal review.
Vacancy discovery	Required	Official ATS, licensed aggregators, direct employer feeds and permitted career pages.
Vacancy trust and moderation	Required	Automated checks, hard blocks, human review and appeals.
Company intelligence	Required progressively	Verified facts first; employee and candidate experience grows with first-party data.
Salary intelligence	Required progressively	Employer-disclosed, government, licensed and first-party observations with labels.
Autonomous applications	Required for supported sources	No promise of universal coverage.
Mailbox tracking	Later release	Gmail/Outlook OAuth, response classification and action items.
Billing and regional pricing	Later release	INR, USD, EUR, entitlements and credits.

9. Role Selection and Automation Authorization

9.1 Role output

- Primary role
- Strong matches
- Related roles
- Stretch roles
- Manual role search with aliases, abbreviations and spelling tolerance

9.2 Authorization screen

After target roles are selected, the product presents a single clear authorization step. It explains that JobBeacon will discover, verify and apply automatically using verified candidate facts and stored global constraints. This is not a campaign wizard.

- Selected target roles and locations derived from profile preferences.
- Global exclusions and hard limits that materially affect eligibility.
- Clear statement that no vacancy-by-vacancy approval is required.
- Pause, resume and stop controls.
- Action-required exception policy.
- Consent and audit timestamp.
- Ability to modify roles later without redoing completed applications.

9.3 Prohibited interaction patterns

- Mandatory job shortlist

- Per-job Approve or Apply button
- Add to campaign as the primary job action
- Required daily-limit wizard before automation
- Repeated consent on every application
- Hidden automatic behavior

10. Vacancy Source Strategy and Policy

There is no universal API key for all job platforms. JobBeacon must use a source capability registry and support multiple acquisition and submission methods. Source access is enabled only after technical, contractual and legal review.

10.1 Preferred source hierarchy

Tier	Source type	Primary use	Application capability
1	Official employer ATS job-board APIs	High-confidence published vacancies	Automatic only with employer/ATS submission authorization or permitted hosted flow.
2	Licensed job aggregator APIs	Broad discovery and salary/location coverage	Usually redirect or source-specific; must follow licence and attribution.
3	Direct employer feeds and partnerships	Verified employer inventory	Best path for full discovery and direct application authorization.
4	Permitted employer career pages	Public vacancy discovery	Browser automation only when terms and controls allow it.
5	Government job APIs and company registries	Official jobs and employer identity	Source-specific application behavior.

10.2 Initial source matrix

Source	Discovery	Application	Product requirement
Greenhouse Job Board API	Public GET endpoints do not require authentication. [S1]	Application POST requires Basic Auth/API credentials controlled by the employer and should be proxied server-side. [S1]	Use public discovery broadly; enable direct application only with authorized credentials or permitted hosted flow.
Lever Postings API	Published jobs are available through the public postings API. [S2]	POST application requires an employer-generated API key and rate-limit handling. [S2]	Use for discovery; direct submission only with employer authorization.
Adzuna	Requires app_id and app_key; supports jobs and salary/employment data. [S3]	Not a universal application endpoint.	Obey attribution, licence, rate and retention requirements.
Jooble	Requires a REST API key and returns job title, company, salary, source link and updated time. [S4]	Normally links back to source.	Use as discovery input; re-verify authoritative source before application.
USAJOBS	Search API requires API key and identifying headers. [S5]	Follow federal application route and requirements.	Support as an official US federal source.

Source	Discovery	Application	Product requirement
Direct employer feeds	Employer-authorized XML/JSON/webhooks/SFTP.	May include direct application API or ATS credentials.	Priority partnership model.

10.3 Prohibited or restricted sources

Source	Restriction	JobBeacon policy
LinkedIn	User Agreement prohibits scraping/copying with scripts or robots, and unauthorized automated access/activity. [S6][S7]	No scraping, copied cookies, stealth browser automation or bot applications without explicit written permission.
Indeed	API and platform use is limited to approved integrations; scraping, database creation and algorithmic calls outside approved use are restricted. [S8]	Use only through an approved API, plugin or written partnership.
Employee-review platforms	Terms may prohibit scraping and republishing.	Use licensed APIs/partnerships or build first-party review data.

10.4 Source policy registry

Field	Example
source_code	greenhouse_public
discovery_allowed	true
storage_allowed	true with retention policy
display_allowed	true with attribution
automated_application_allowed	conditional: employer credential required
authentication_method	none for GET; server-held credential for POST
rate_limit	provider-specific
countries	provider coverage
policy_version	terms/API documentation version
last_legal_review_at	timestamp
kill_switch	enabled/disabled

11. Vacancy Ingestion and Normalization

11.1 Ingestion pipeline

14. Fetch source records under the source policy.
15. Store raw source snapshot and retrieval metadata.
16. Normalize title, company, location, work mode, employment type, salary and description.
17. Resolve canonical company and legal entity.

18. Resolve canonical role taxonomy.
19. Generate stable source and cross-source fingerprints.
20. Deduplicate exact and near-duplicate postings.
21. Recheck source freshness before matching and before submission.
22. Version changes without overwriting historical evidence.

11.2 Canonical vacancy fields

Domain	Required fields
Identity	canonical_vacancy_id, source_vacancy_id, source_code, authoritative_url
Role	raw_title, canonical_role_id, seniority, function, skill requirements
Company	company_id, legal_entity_id, displayed_name, domain, career_domain
Location	country, region, city, remote/hybrid/on-site, timezone constraints
Compensation	currency, min, max, interval, disclosed/estimated/source
Lifecycle	published_at, discovered_at, last_seen_at, expires_at, status
Trust	source confidence, vacancy trust status, hard-block reasons
Application	application channel, required fields, action-required capabilities

11.3 Deduplication

- Exact source ID match
- Canonical URL match
- Company + normalized title + location + publish-window fingerprint
- Semantic description similarity
- Cross-source employer career-page confirmation
- Repost version relationship rather than false new vacancy

12. Vacancy Verification and Trust Scoring

12.1 Status model

Status	Meaning	Automatic application
VERIFIED	Passed critical source, employer, vacancy and safety checks.	Allowed when eligible and technically supported.
VERIFIED_INCOMPLETE	Legitimate but some non-critical fields are missing.	Allowed subject to candidate hard constraints.
UNDER_REVIEW	Conflicting, unusual or incomplete evidence.	No.
FLAGGED	One or more serious risk signals.	No.

Status	Meaning	Automatic application
BLOCKED	Confirmed fake, phishing, payment request, unauthorized source or prohibited content.	Never.
EXPIRED_REMOVED	No longer active on authoritative source.	Never.
ACTION_REQUIRED	Legitimate, but submission needs unavoidable candidate input.	Paused for that application.

12.2 Trust score

Dimension	Weight	Examples
Employer identity	20	Legal entity, official domain, registry status
Authoritative source	20	Official career page, ATS or authorized feed
URL and application integrity	15	TLS, redirect chain, domain match, no phishing
Freshness and availability	10	Still published, recent last-seen check
Content consistency	10	Role/company/location consistency across sources
Candidate and moderator history	10	Reports, prior decisions, appeal outcomes
Salary plausibility	5	Range plausible for role/location/seniority
Scam and prohibited-content signals	10	Payment, MLM, identity collection, cheque/crypto

The score is explainable but not absolute. Hard-block rules override the numeric result.

12.3 Hard-block reasons

- PAYMENT_OR_FEE_REQUEST
- PHISHING_OR_MALWARE_REDIRECT
- COMPANY_IMPERSONATION
- UNAUTHORIZED_SOURCE_ACCESS
- MLM_OR_PYRAMID_RISK
- PREMATURE_BANK_OR_GOVERNMENT_ID_REQUEST
- VACANCY_REMOVED
- PROHIBITED_OR_ILLEGAL_REQUIREMENT
- DOMAIN_MISMATCH_WITH_NO_EXPLANATION
- CONFIRMED_MODERATOR_BLOCK

12.4 Positive reason codes

- OFFICIAL_CAREER_PAGE_CONFIRMED
- ATS_POSTING_CONFIRMED
- COMPANY_REGISTRY_CONFIRMED

- CORPORATE_DOMAIN_CONFIRMED
- RECENT_SOURCE_RECHECK_PASSED
- SALARY_EMPLOYER_DISCLOSED
- PRIOR_TRUSTED_EMPLOYER_HISTORY

13. Fake-Job Detection and Moderation

13.1 Moderation layers

Layer	Responsibilities	Output
Automated	Rules, URL analysis, domain checks, duplicate/repost detection, content models, salary anomaly and source-policy enforcement.	Verify, quarantine or hard-block with reason codes.
Human moderator	Review high-risk, conflicting, appealed or high-impact cases.	Clear, flag, block, request evidence or escalate.
Candidate/community reports	Report fake job, payment request, impersonation, salary mismatch, expired job or misleading remote status.	Create a case with abuse controls and evidence.
Employer claim/appeal	Correct objective facts and appeal a block.	Evidence-based correction or upheld decision.

13.2 Moderator case workflow

23. Case created by rule, report or appeal.
24. Severity and candidate-exposure impact assigned.
25. Evidence snapshot frozen.
26. Related company, source and duplicate cases linked.
27. Moderator reviews reason codes and raw evidence.
28. Decision recorded with policy version and rationale.
29. Affected pending applications cancelled or released.
30. Employer/candidate notified where appropriate.
31. Appeal window and reviewer separation enforced.
32. Final decision retained in audit history.

13.3 Moderation SLAs

Severity	Examples	Initial response target
Critical	Payment scam, phishing, malware, impersonation with active applications	15 minutes automated containment; human review within 4 hours
High	Sensitive-data harvesting, repeated credible reports	Contain immediately; review within 12 hours
Medium	Conflicting company facts, misleading salary/remote status	Review within 2 business days
Low	Expired listing, typo, duplicate	Review or automated closure within 5 business days

13.4 Review integrity

- Moderator cannot review their own prior decision on appeal.
- Employer payment or subscription status must not influence trust decisions.
- Negative reviews are removed only for policy violations, not because they are unfavorable.
- Candidate reports are rate-limited and abuse-scored.
- All policy and model versions are recorded.

14. Company Profile and Company Intelligence

Company profiles separate objective verification from subjective experience. JobBeacon must not compress legitimacy, work-life balance, salary and candidate experience into one unexplained star rating.

14.1 Verified company facts

- Legal and trading names
- Registration number and status
- Official website and verified domains
- Verified career domains and ATS boards
- Headquarters and operating countries
- Industry and founded year
- Employee-size range
- Public/private status where known
- Official social links
- Employer-claimed profile status
- Active verified vacancy count

For India, the Ministry of Corporate Affairs company master data is available through the Government Open Data platform and includes CIN, company status, registration and capital fields. [S9] For UK companies, Companies House provides a REST API requiring authentication credentials. [S10] For US public companies, SEC data APIs provide company submissions and XBRL data without API keys. [S11]

14.2 Separate company scores

Dimension	Source	Display rule
Company Verification	Registries, official domains, employer evidence	Verified / Partially verified / Unverified / Blocked
Vacancy Trust	JobBeacon vacancy decisions	Percentage plus reason summary and sample count
Work-Life Balance	Verified employee reviews	1–5 only above minimum sample threshold
Compensation & Benefits	Reviews, disclosed salaries and benchmarks	Separate employee rating and market competitiveness
Management & Culture	Verified employee reviews	1–5 with sample size and recency
Career Growth	Verified employee reviews	1–5 with sample size and recency
Candidate Experience	JobBeacon application/interview outcomes	Response rate, time-to-response, interview experience
Application Integrity	Posting vs actual process consistency	Evidence-backed score and issue rate

14.3 First-party review system

JobBeacon’s long-term company intelligence should rely primarily on first-party verified reviews rather than scraped third-party review platforms.

- Anonymous public display with internal relationship-verification status.
- Verification through corporate email, redacted employment document, verified offer/interview evidence or consented mailbox evidence.
- One review per user per company within a defined period.
- No employer incentives, coercion or review gating.
- Duplicate-language, brigading, identity and account-risk checks.
- Human moderation and employer response rights.
- Appeal and correction workflow for factual claims.

14.4 Rating publication thresholds

Review count	UI behavior
0–4	Show “Insufficient verified responses”; no score.
5–9	Show directional score with low-confidence label.
10–19	Show score with sample count and confidence indicator.
20+	Show prominent dimension score, distribution and trend.

15. Salary Intelligence

15.1 Source hierarchy

Priority	Salary type	Example label
1	Employer disclosed in authoritative job posting	Employer disclosed • Updated 24 Jul 2026
2	Government or official occupational benchmark	Government benchmark • Role and region
3	Licensed market data or API estimate	Licensed market estimate • Provider attribution
4	JobBeacon verified employee/offer observation	JobBeacon verified observation • Aggregated
5	JobBeacon model estimate	Model estimate • Confidence and input basis

Adzuna’s API requires an app_id and app_key, provides jobs and salary/employment endpoints, and has attribution and terms requirements. [S3][S12]

15.2 Salary display rules

- Always show currency and interval.
- Never mix annual, monthly and hourly amounts without normalization.

- Label disclosed, benchmark, licensed, verified observation and model estimate distinctly.
- Show updated date and geographic scope.
- Do not publish individual compensation data.
- Apply minimum sample and privacy thresholds to JobBeacon aggregates.
- Display an uncertainty range for model estimates.
- Flag implausible salary as a trust signal, not automatic proof of fraud.

15.3 Regional currency

Region	Candidate display	Storage
India	INR	Original currency + normalized analytical currency
United States	USD	Original currency + normalized analytical currency
Europe	EUR where appropriate; preserve local currency when required	Original currency + EUR analytical value

16. Autonomous Application Engine

16.1 Eligibility gate

An application may proceed only when all mandatory gates pass.

Gate	Pass requirement
Source policy	Discovery and automated application are allowed for this channel.
Vacancy trust	Status is VERIFIED or approved VERIFIED_INCOMPLETE.
Candidate eligibility	Work authorization, location, seniority and hard constraints pass.
Verified facts	Required answers are supported by confirmed candidate data.
Application support	Portal fields and attachments are supported.
Consent and privacy	Required processing and submission consent exists.
Rate and abuse controls	Source and candidate limits permit execution.
Idempotency	No prior successful or active application exists for the same canonical vacancy.

16.2 Application channels

Channel	Behavior
Authorized ATS API	Server submits using employer/source credentials and handles validation/rate limits.
Permitted hosted form	Controlled browser worker fills and submits the public form.
Employer direct API/feed	Use the contracted interface and evidence response.

Channel	Behavior
Redirect-only source	Follow to authoritative employer/ATS destination, then re-evaluate policy.
Unsupported/restricted	Skip or mark unsupported; do not bypass restrictions.

16.3 ATS resume generation

- Use only confirmed master-profile facts.
- Select evidence relevant to the vacancy.
- Never add unsupported skills, employers, dates or metrics.
- Record source facts, template version, model/prompt version and output hash.
- Run factuality, formatting and ATS quality checks.
- Preserve immutable generated version and application linkage.

16.4 Worker lifecycle

33. Lease one application task atomically.
34. Recheck vacancy freshness and trust.
35. Load verified candidate facts and source policy.
36. Generate or select application assets.
37. Complete supported questions.
38. Pause on unknown or sensitive questions.
39. Submit idempotently.
40. Capture confirmation, timestamps, screenshots/receipts where permitted.
41. Classify success, failure, retry or action required.
42. Release lease and schedule retry safely.

16.5 Candidate UI

- Autonomous operation status
- Selected roles
- Discovered / verified / applied / blocked counts
- Recent applications and evidence
- Response and interview events
- Action-required queue
- Pause, resume, stop and change-role controls
- No mandatory per-job approval

17. Action-Required Exceptions

Exception	System behavior	Candidate request
CAPTCHA	Pause only the affected application and preserve state.	Complete challenge in a secure handoff.
OTP or email code	Wait with expiry timer and safe retry.	Provide or complete OTP through permitted flow.
Unknown legal/sensitive question	Do not infer or answer.	Provide one answer and optionally save

Exception	System behavior	Candidate request
		as verified preference.
Missing verified fact	Pause and identify exact missing data.	Confirm or add the fact.
Assessment/video/interview	Mark external action required.	Complete assessment or schedule interview.
Unsupported portal	Skip, route to manual queue or await adapter support.	Optional manual completion; not routine.
Payment or financial request	Block and create moderation case.	No payment request is forwarded as normal action.

17.1 Action-required UX

- Explain why automation stopped.
- Show affected company and vacancy.
- State what JobBeacon has already saved.
- Provide one primary action.
- Show deadline or expiry when relevant.
- Allow skip where safe.
- Resume automatically after successful completion.
- Never expose secrets or raw provider tokens.

18. Candidate Experience Requirements

18.1 Primary navigation

- Overview
- Profile
- Resumes
- Target Roles
- Opportunities
- Applications
- Responses
- Action Required
- Company Intelligence
- Security

18.2 Opportunities screen

This screen shows opportunities discovered and verified by JobBeacon. It is an evidence and monitoring surface, not a manual job board.

- Verification status and trust explanation
- Company profile summary
- Match and eligibility explanation
- Salary source and confidence
- Freshness and source
- Automatic application status

- Blocked or action-required reason
- Evidence link
- No primary Save, Add to campaign or Apply action

18.3 Company profile screen

- Verified company identity
- Trust and vacancy history
- Separate employee-experience dimensions
- Salary competitiveness
- Candidate response and interview metrics
- Active verified vacancies
- Review count, recency and confidence
- Employer response and correction status

18.4 Accessibility and responsive requirements

- WCAG 2.2 AA target
- Keyboard-complete workflows
- Persistent labels and visible focus
- No color-only status
- 320, 375, 390, 412, 768, 1024 and 1440 px checks
- 200% zoom and large-text support
- Reduced-motion support
- No horizontal page scroll
- Live-region announcements for application and action-required status

19. Moderator and Admin Experience

19.1 Moderator dashboard

- Risk queue by severity and candidate exposure
- Vacancy and company evidence timeline
- Source-policy status
- Automated reason codes and score breakdown
- Duplicate and related-case graph
- Candidate report history
- Employer claim and appeal evidence
- Clear/flag/block/request-info/escalate actions
- Decision rationale required
- SLA timer and ownership

19.2 Source management

- Enable/disable discovery and application separately
- Credential references via secret manager
- Rate limits and retry policy
- Attribution requirements
- Data retention and display rights

- Country coverage
- Terms version and legal review date
- Emergency kill switch
- Source health and error metrics

19.3 Trust configuration

- Rule thresholds
- Hard blocks
- Model versions
- Review sampling
- False-positive analysis
- Moderator override permissions
- Appeal policy
- Company-review publication thresholds

20. Employer Claim, Correction and Appeals

20.1 Claiming a profile

- Corporate-domain verification
- Authority evidence
- Legal entity matching
- Named representative and role
- MFA-protected employer account
- Audit and annual reverification

20.2 Employer capabilities

- Correct objective company facts with evidence
- Register career domains and ATS boards
- Provide authorized vacancy feeds and application credentials
- Respond publicly to reviews
- Appeal a vacancy or company block
- View aggregated trust and candidate-experience insights
- Cannot identify anonymous reviewers or delete valid negative reviews

20.3 Appeals

- Separate reviewer from original decision
- Evidence submission deadline
- Temporary status rules during appeal
- Decision rationale and policy citation
- Further escalation for high-impact cases
- Full immutable history

21. Data Model

21.1 Core entities

Domain	Entities
Candidate	candidate_profiles, candidate_preferences, candidate_selected_roles, candidate_exclusions
Resume	resume_documents, resume_versions, resume_parse_runs, extracted_facts, fact_confirmations
Company	companies, company_legal_entities, company_domains, company_registry_records, company_profiles
Company experience	company_reviews, company_review_verifications, company_rating_snapshots, employer_responses
Salary	salary_observations, salary_benchmarks, salary_estimates
Sources	vacancy_sources, source_policies, source_credentials, source_health_events
Vacancies	vacancies, vacancy_versions, vacancy_source_records, vacancy_fingerprints
Trust	vacancy_trust_scores, vacancy_flags, vacancy_evidence, moderation_cases, moderation_decisions, appeals
Applications	automation_authorizations, application_plans, application_attempts, application_evidence, action_required_events
Responses	mailbox_connections, messages, response_classifications, interviews, candidate_action_items
Audit	audit_events, model_executions, policy_versions, security_events

21.2 Important constraints

- Candidate-owned data uses RLS and candidate-specific storage paths.
- Raw vacancy snapshots are immutable and versioned.
- One canonical vacancy may map to multiple source records.
- Trust decisions reference evidence and policy/model versions.
- Application attempts are idempotent per candidate + canonical vacancy.
- Moderation decisions cannot be deleted; corrections create new versions.
- Review public identity remains anonymous while verification metadata is protected.
- Salary observations are aggregated under privacy thresholds.

22. API and Event Contracts

22.1 API domains

- Authentication and sessions
- Resume upload, parsing and fact confirmation
- Role suggestions and selections
- Automation authorization and stop controls
- Vacancy discovery and status

- Vacancy trust and evidence
- Company profiles, ratings and reviews
- Moderation and appeals
- Application attempts and evidence
- Action-required responses
- Mailbox and response tracking
- Admin source policy and health

22.2 Core events

- candidate.roles_selected.v1
- automation.authorized.v1
- vacancy.discovered.v1
- vacancy.normalized.v1
- vacancy.verified.v1
- vacancy.flagged.v1
- vacancy.blocked.v1
- company.verified.v1
- application.planned.v1
- application.action_required.v1
- application.submitted.v1
- application.failed.v1
- response.received.v1
- interview.detected.v1
- moderation.decision_recorded.v1
- employer.appeal_submitted.v1

22.3 Contract standards

- Versioned schemas
- Typed validation
- Correlation and causation IDs
- Idempotency keys
- Standardized errors
- Redacted logs
- Audit event on material action
- Retry classification
- Source and policy version
- Model and prompt version where AI is involved

23. System Architecture

Component	Responsibility
Candidate web application	Onboarding, role selection, monitoring, evidence, action-required and controls.
Admin/moderation application	Source policy, case review, appeals, employer claims and trust analytics.

Component	Responsibility
Supabase/PostgreSQL	Candidate data, source registry, vacancies, trust, applications, reviews and audit.
Private object storage	Resumes, generated variants and permitted evidence.
Ingestion workers	Fetch allowed sources, normalize, deduplicate and version.
Trust workers	Company resolution, URL checks, rules and model scoring.
Application workers	ATS/API/browser adapters, idempotent submission and evidence.
Mailbox workers	OAuth ingestion, classification and application matching.
Queue/leasing layer	PostgreSQL-backed tasks, leases, retries and dead-letter handling.
Secret manager	Source and employer credentials; no secrets in browser or repository.
Observability	Metrics, logs, traces, alerts, SLOs and security monitoring.

23.1 Adapter architecture

- DiscoveryAdapter
- VacancyNormalizer
- CompanyResolver
- TrustEvaluator
- EligibilityEvaluator
- ResumeVariantGenerator
- ApplicationAdapter
- EvidenceCollector
- ResponseClassifier
- ModerationProvider

23.2 Capability checks

The application orchestrator must evaluate source capabilities at runtime. A source that allows discovery but not automated submission may still provide a vacancy, but the system must switch to an authorized destination or classify it as unsupported rather than silently violating the source policy.

24. AI/ML Requirements and Safeguards

24.1 Permitted AI use

- Resume extraction and normalization
- Role suggestion and evidence explanation
- Vacancy classification and canonicalization
- Duplicate and semantic similarity
- Scam-language and anomaly detection
- Eligibility assistance under deterministic hard rules
- ATS alignment and truthful summarization

- Response classification
- Review abuse detection

24.2 Prohibited AI behavior

- Inventing candidate facts
- Answering unknown legal or sensitive questions
- Overriding source policy
- Overriding hard-block moderation rules
- Publishing unexplained company or trust scores
- Using protected attributes for unlawful discrimination
- Automatically removing a review based only on negative sentiment
- Claiming certainty when evidence is incomplete

24.3 Evidence and quality

- Record model, prompt and schema version.
- Reject invalid structured output.
- Preserve supporting facts and evidence.
- Use confidence and bounded repair.
- Human review for high-risk moderation ambiguity.
- Monitor drift, false positives and false negatives.
- Maintain test fixtures for fake jobs, legitimate edge cases and international formats.

25. Security, Privacy and Compliance

25.1 Candidate security

- RLS and cross-candidate isolation
- Private resume storage and short-lived signed URLs
- MFA/AAL2 for sensitive controls
- Session revocation and security history
- No service-role or source secrets in browser
- Encrypted secrets and least privilege
- Consent and purpose limitation
- Deletion and retention controls
- Redacted logs and evidence

25.2 Source and automation compliance

- Source-specific terms review before activation
- No bypass of robots, access controls or CAPTCHA
- No LinkedIn/Indeed scraping without explicit permission
- Attribution and retention compliance
- Rate-limit and user-agent compliance
- Kill switch by source, country and capability
- Legal review date stored with source policy
- Automated application only through authorized channels

25.3 Review privacy

- Public anonymity
- Internal verification separated from public profile
- Minimum sample thresholds
- No publication of individual salary or employment documents
- Employer cannot access reviewer identity
- Abuse reports and legal requests handled through controlled workflow

25.4 Company registry data

Company registry integrations are used for objective verification, not as a complete quality rating. Registry data may be incomplete, delayed or unavailable for certain jurisdictions and entities.

26. Analytics and KPIs

26.1 North-star metric

Verified autonomous application outcomes

Number and rate of source-authorized, vacancy-verified, candidate-eligible applications submitted with valid evidence and without routine candidate assistance.

26.2 Candidate metrics

- Time from role selection to first verified application
- Verified applications per active candidate
- Action-required events per 100 attempts
- Candidate minutes required per 100 applications
- Interview and response rates
- Pause/stop rate
- Candidate trust and satisfaction
- False application or duplicate complaint rate

26.3 Trust metrics

- Vacancies blocked before candidate exposure
- Fake-job precision and recall
- Moderator overturn rate
- Employer appeal uphold/overturn rate
- Time to contain critical scam
- Source-specific stale and duplicate rate
- Company verification coverage
- Review verification and abuse rates

26.4 Operational metrics

- Source fetch success and latency

- Application adapter success by source
- Retry and dead-letter rate
- Evidence completeness
- Mailbox matching accuracy
- Cost per verified application
- API quota utilization
- Model cost and latency

27. Regionalization and Monetization

27.1 Markets

The product launches globally in principle but source coverage, application automation, company registries and legal controls are enabled market by market. Initial product and pricing support INR, USD and EUR.

27.2 Plan limits

- Active target roles
- Verified applications per month
- Premium source access
- ATS resume variants
- Mailbox connections
- Company intelligence depth
- Priority action-required support
- Historical analytics and exports

27.3 Safety limits vs pricing limits

Commercial entitlements never override source rate limits, trust blocks, candidate eligibility, legal restrictions or safety controls. “Unlimited” must not be used when applications are constrained by plans, source policies and eligibility.

28. Rollout Plan

Release	Scope	Exit gate
R0 — Interactive preview	Synthetic role selection, discovery, moderation, company profile and simulated applications.	Founder product review; no production claims.
R1 — Identity and verified profile	Auth, MFA/AAL2, private resume, facts and role selection.	Security, RLS, storage and founder acceptance.
R2 — Discovery foundation	Greenhouse, Lever, USAJOBS, one licensed aggregator, normalization and dedupe.	Source-policy review, freshness and duplicate accuracy.
R3 — Trust and moderation	Company resolution, trust scoring, fake-job blocks, moderator console and reports.	Precision/recall targets and critical scam containment.
R4 — Authorized automatic applications	Selected employer/ATS partners and permitted hosted flows.	Submission success, idempotency, evidence and action-required safety.
R5 — Company and salary intelligence	Verified facts, first-party reviews, salary layers and employer claim.	Privacy thresholds, moderation and appeal acceptance.

Release	Scope	Exit gate
R6 — Mailbox and responses	Gmail/Outlook OAuth, response classification and interviews.	Security and matching accuracy.
R7 — Billing and regional scale	Plans, credits, INR/USD/EUR, entitlements and reconciliation.	Financial controls and production certification.

29. Acceptance Criteria

29.1 Candidate autonomy

- After verified role selection and authorization, a candidate can receive verified automatic applications without job-by-job approval.
- Candidate can pause, resume or stop.
- Changing roles affects future discovery without duplicating prior applications.
- Routine monitoring is optional.
- Only defined action-required exceptions interrupt the candidate.

29.2 Source compliance

- Every active source has a current source-policy record.
- Discovery and application permissions are evaluated separately.
- Restricted platforms are not scraped or automated.
- Credentials remain server-side.
- Rate, attribution and retention requirements are tested.

29.3 Trust and moderation

- Critical scam signals block immediately.
- Every trust decision has reason codes and evidence.
- Flagged jobs do not enter automatic application.
- Moderator decisions and appeals are auditable.
- False-positive and false-negative test suites pass agreed thresholds.

29.4 Company intelligence

- Objective facts are source-labelled.
- Employee experience, compensation and candidate experience are separate.
- Low-sample ratings are not published as reliable scores.
- Employer corrections require evidence.
- Negative reviews cannot be removed merely because they are unfavorable.

29.5 Application engine

- Applications are idempotent.
- Vacancy is rechecked before submission.
- Generated resume is factual and linked to confirmed facts.
- Submission evidence is complete.
- Unknown sensitive questions are escalated, not guessed.

- Unsupported or prohibited channels are skipped.

29.6 Quality and release

- Typecheck, lint, unit, integration, browser and accessibility tests pass.
- Clean migration reset passes.
- Candidate A/B isolation passes.
- Source and application adapters have fixture tests.
- Deployment SHA equals tested SHA and artifact SHA.
- Founder acceptance passes.

30. Risks and Mitigations

Risk	Impact	Mitigation
Source terms change	Discovery or application becomes unauthorized.	Version source policy, scheduled legal review, automated kill switch.
Fake-job false negative	Candidate exposed to fraud.	Hard rules, layered verification, recheck before submission, human review.
False positive blocks legitimate job	Lost opportunity and employer dispute.	Explainable reasons, appeals, model monitoring and separate reviewer.
Automation submits wrong answer	Candidate harm or legal risk.	Verified facts only, deterministic validation, action-required escalation.
Company rating manipulation	Misleading company intelligence.	Relationship verification, abuse detection, sample thresholds and moderation.
Salary data misleading	Bad candidate decisions.	Source labels, confidence, recency and separate disclosed/estimated values.
Portal changes	Application failures.	Adapter versioning, health checks, retry classification and kill switch.
Credential leakage	Source or candidate compromise.	Secret manager, least privilege, redaction and no browser exposure.
Over-application or spam	Employer/platform harm.	Eligibility, quality, source limits, candidate hard constraints and abuse controls.
Global legal variation	Compliance exposure.	Country-level capability flags, counsel review and staged rollout.

31. Open Product Decisions

Decision	Founder choice required
Automation authorization wording	Exact consent copy and whether one-time authorization is role-specific or global.
Global default limits	Whether candidates may set optional daily/total limits or only pause/stop.
Unsupported application handling	Skip automatically, offer manual completion, or queue for future adapter support.
Review verification	Accepted evidence types by region and privacy burden.

Decision	Founder choice required
Rating thresholds	Minimum samples and weighting for public display.
Composite company score	Whether to omit entirely or offer an optional transparent summary.
Source rollout order	Prioritize ATS breadth, India coverage, US coverage or employer partnerships.
Moderator staffing	In-house, outsourced or hybrid for launch.
Appeal SLAs	Paid/unpaid employers must receive the same trust treatment; define response times.
Candidate exclusions	Default exclusions for staffing agencies, contract roles, relocation, salary and sensitive sectors.

32. Official Source References

These sources are implementation inputs, not permanent permission. Source terms and API documentation must be re-reviewed before launch and periodically afterward.

[S1] Greenhouse Job Board API: <https://developers.greenhouse.io/job-board.html>

[S2] Lever Postings API: <https://github.com/lever/postings-api>

[S3] Adzuna API Overview: <https://developer.adzuna.com/overview>

[S4] Jooble REST API: <https://jooble.org/api/about>

[S5] USAJOBS API Quick Start: <https://developer.usajobs.gov/general/quick-start>

[S6] LinkedIn User Agreement: <https://www.linkedin.com/legal/user-agreement>

[S7] LinkedIn Crawling Terms: <https://www.linkedin.com/legal/crawling-terms>

[S8] Indeed Developer Agreement: <https://docs.indeed.com/legal-terms/developer-agreement>

[S9] India Company Master Data — Ministry of Corporate Affairs / data.gov.in: <https://data.gov.in/catalog/company-master-data>

[S10] Companies House API — Get Started: <https://developer.company-information.service.gov.uk/get-started>

[S11] SEC EDGAR Data APIs: <https://www.sec.gov/search-filings/edgar-application-programming-interfaces>

[S12] Adzuna API Terms of Service: https://developer.adzuna.com/docs/terms_of_service

Founder Approval Checklist

- Approve the autonomy model: role selection is the final routine candidate decision before automatic discovery and application.
- Approve the source hierarchy and restricted-platform policy.
- Approve the vacancy trust statuses, hard blocks and moderation model.
- Approve separate company intelligence dimensions and review thresholds.
- Approve salary source labels and confidence treatment.
- Approve the staged rollout plan and initial source priorities.
- Approve the open product decisions or assign owners and deadlines.

FINAL PRODUCT BOUNDARY

Highly autonomous. Never unauthorized.

Release rule

JobBeacon applies to a vacancy automatically only when the source permits the behavior, the company and vacancy are trusted, the candidate is eligible, the required answers are supported by verified facts, and the application channel is technically supported.

Founder approval converts this draft into the implementation authority for sourcing, moderation, company intelligence and autonomous applications.